

Magnetic & Soft Medical Robotics

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Design and experimental validation of miniaturized magnetic and compliant robotic systems for minimally invasive imaging and intervention.

- Miniaturized imaging catheters
- Magnetic actuation and navigation
- Soft robotic systems

SELECTED OUTPUTS

ICBIR 2026

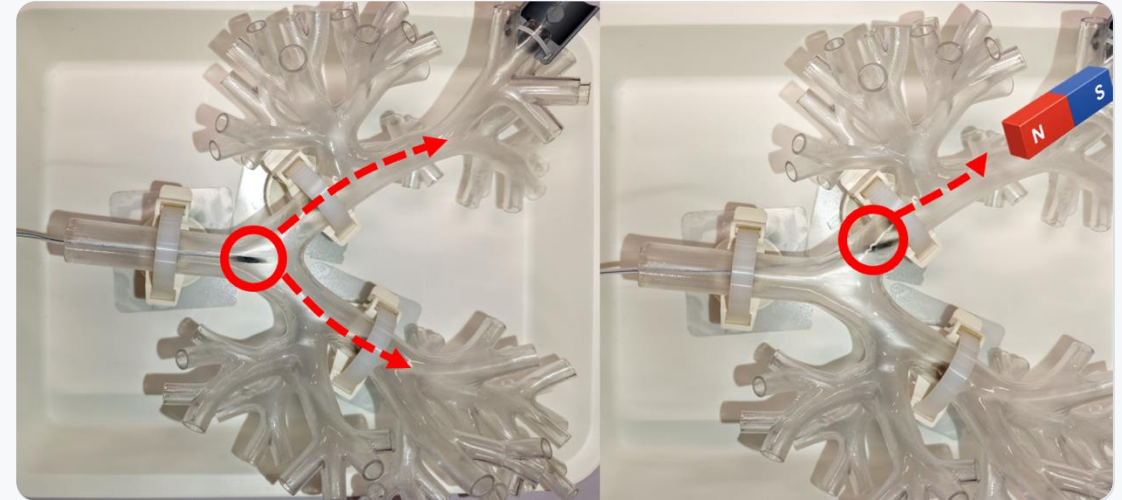
Co-first author

ICRA 2026

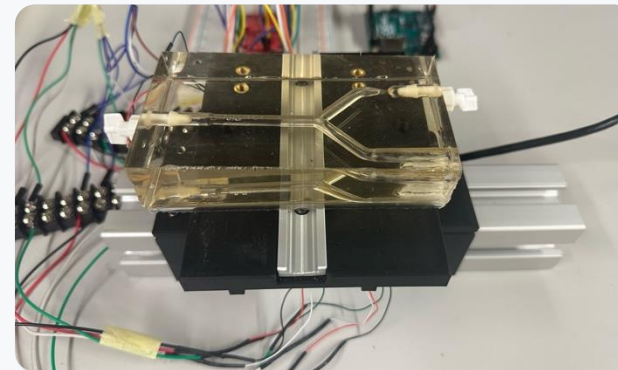
Accepted paper

ETH Zürich MSRL

Semester project · 5.5/6.0



Magnetic branch selection in a transparent bronchial-tree phantom



Quantitative characterization



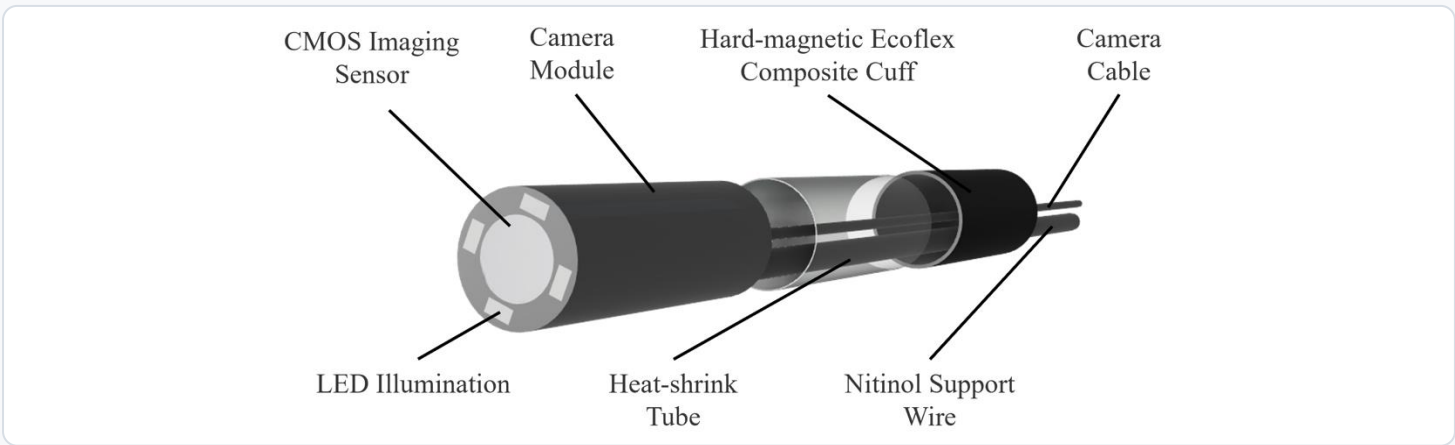
Control · integration · ex vivo validation

Magnetically Steerable Micro-Camera Catheter for Bronchial Navigation

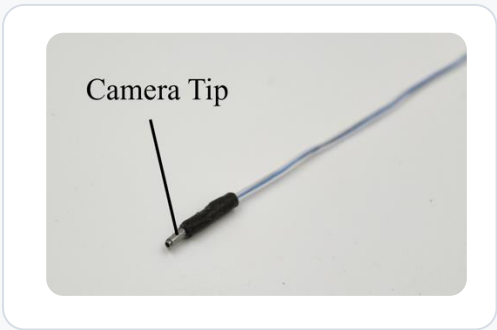
Co-first author · Accepted at ICBIR 2026 · CUHK · Advisor: Prof. Hongliang Ren

ENGINEERING PROBLEM Enable miniaturized forward-view imaging and controllable branch selection inside a bronchial-tree environment.

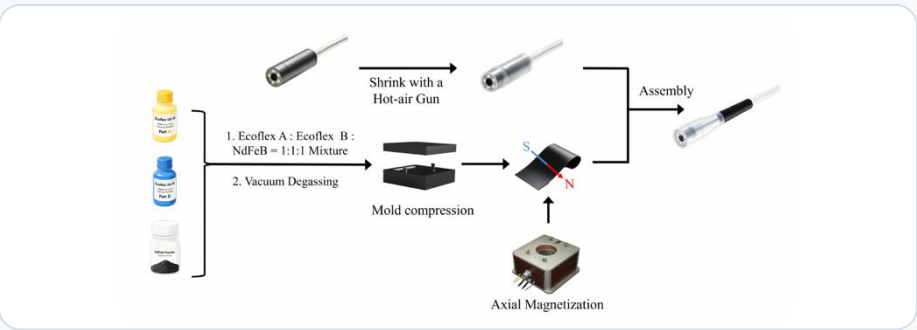
SYSTEM ARCHITECTURE



Camera + LED illumination · nitinol support · heat-shrink encapsulation · hard-magnetic composite cuff



Prototype



Composite preparation → magnetization → distal assembly

MY ROLE

01 DESIGN & FABRICATION

Designed and fabricated the millimeter-scale micro-camera catheter.

02 SYSTEM INTEGRATION

Integrated the camera, illumination, nitinol support, heat-shrink encapsulation, and hard-magnetic composite cuff.

03 EXPERIMENT DEVELOPMENT

Developed the insertion-force and steering-characterization setups.

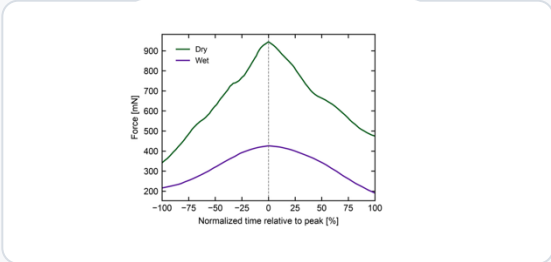
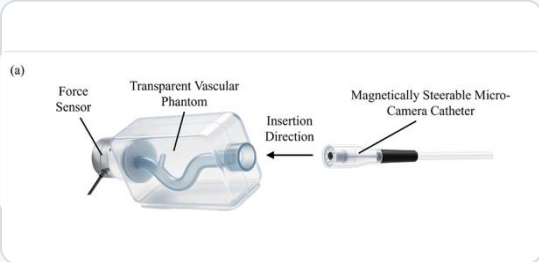
04 VALIDATION & DOCUMENTATION

Conducted bronchial-tree phantom navigation and contributed to manuscript and figure preparation.

Magnetic Steering Enabled Branch Selection While Preserving Forward Imaging

QUANTITATIVE CHARACTERIZATION

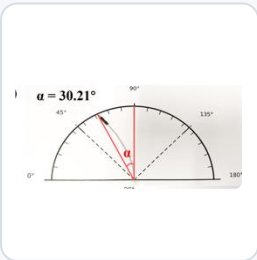
INSERTION FORCE



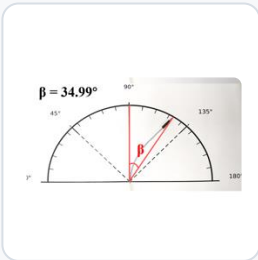
943.1 → 426.0 mN

Dry → saline-wetted · 10 repeated trials per condition

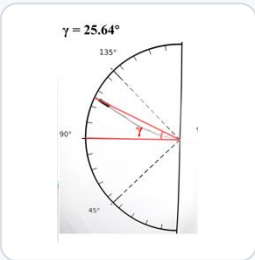
STEERING RANGE



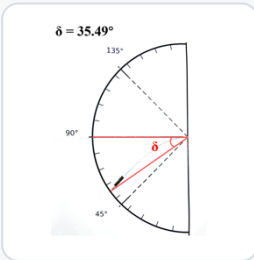
30.2°
left



35.0°
right



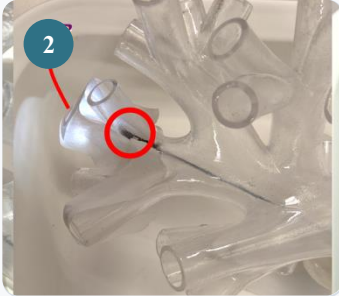
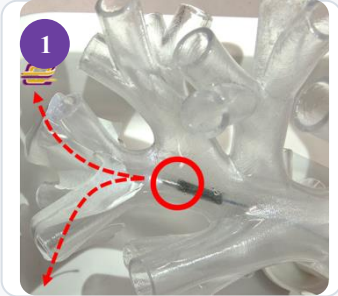
25.6°
up



35.5°
down

NAVIGATION AND IMAGING DEMONSTRATION

BRONCHIAL BIFURCATION SEQUENCE



FORWARD-LOOKING CAMERA VIEWS



Magnetic steering enabled branch-to-branch redirection in a bronchial-tree phantom.

Characterized insertion force and steering behavior.

Demonstrated redirection across bronchial-tree branches.

Maintained forward-looking imaging during benchtop navigation.

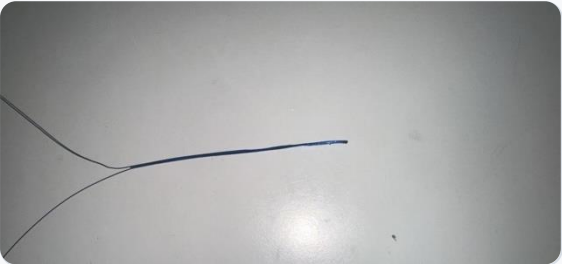
Soft Miniaturized Intraductal Imaging Catheter for Breast Duct Visualization

ETH Zürich · Multi-Scale Robotics Lab · Advisor: Prof. Bradley Nelson · Semester project grade: 5.5/6.0

DESIGN QUESTION

How can catheter diameter and insertion force be reduced while preserving sufficient pushability for intraductal imaging?

1 · DESIGN



0.5 mm and 0.7 mm nitinol supports

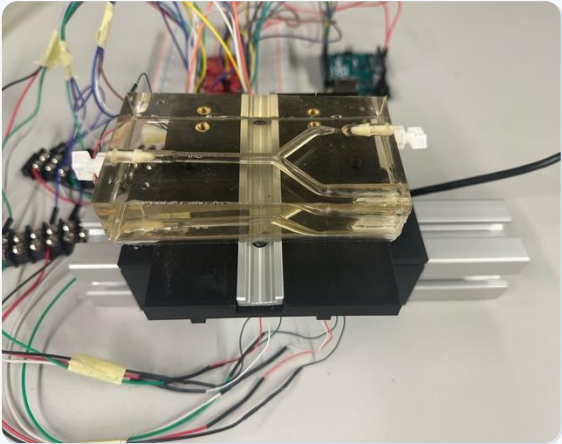


Soft phantom



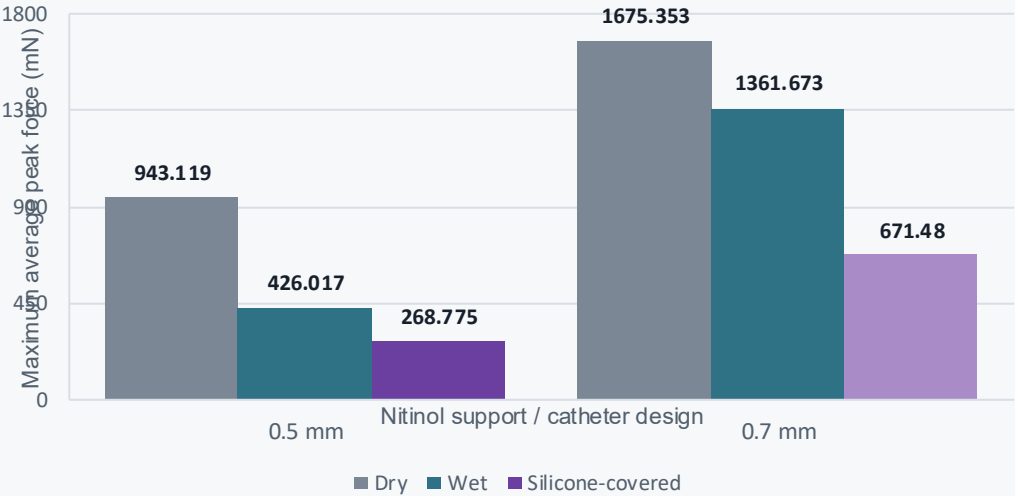
Camera view

2 · MEASUREMENT



Four load cells converted the catheter insertion event into a calibrated insertion-axis force measurement.

3 · RESULT



0.5 mm + silicone covering lowest measured force: 268.8 mN

10 selected trials analyzed per condition (20 manual trials recorded).

MY ROLE

Prototype design and fabrication

Force-platform development and calibration

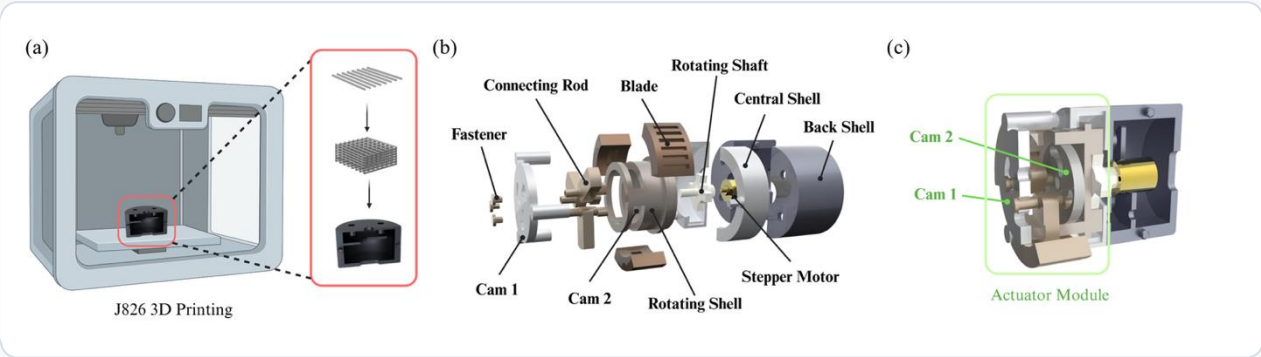
Experimental comparison

Phantom fabrication and imaging validation

Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy

Contributing author · Accepted at ICRA 2026 · CUHK · Advisor: Prof. Hongliang Ren

— A · MECHANISM



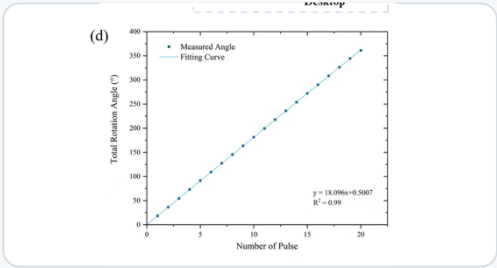
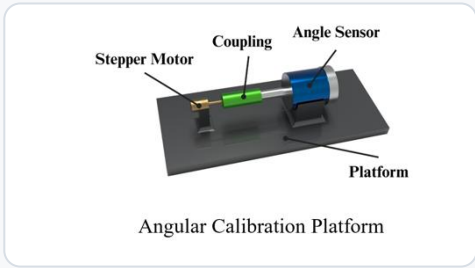
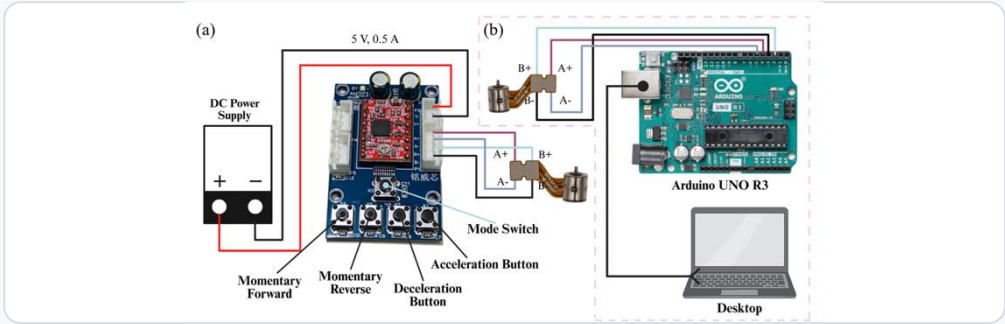
Exploded assembly clarifies the dual-cam actuator, rotating shaft, shell, and stepper-motor integration.

— C · ASSEMBLY AND VALIDATION



Ex vivo system validation

— B · CONTROL AND CALIBRATION



Contribution centered on control, calibration, integration, and validation.

$R^2 = 0.99$

MY CONTRIBUTION

- Developed motor-control electronics and Arduino pulse-control logic.
- Built and calibrated the angular-position measurement platform.
- Contributed to iterative mechanical assembly and system integration.
- Participated in ex vivo experiments and quantitative analysis.